

1. Prove that for any full quadtree Q , we have $n(Q) = \ell(Q) + i(Q)$.

- BASIS STEP: For a full quadtree Q consisting of only a root, $n(Q) = 1$, $\ell(Q) = 1$, and $i(Q) = 0$
 $\therefore n(Q) = \ell(Q) + i(Q)$ holds ($1 = 1 + 0$)
- RECURSIVE STEP: Assume $n(Q_1) = \ell(Q_1) + i(Q_1)$, $n(Q_2) = \ell(Q_2) + i(Q_2)$, $n(Q_3) = \ell(Q_3) + i(Q_3)$, and $n(Q_4) = \ell(Q_4) + i(Q_4)$, when Q_1, Q_2, Q_3 , and Q_4 are pairwise disjoint full quadtrees, and r is a vertex not belonging to them, (Q_1, Q_2, r, Q_3, Q_4) , then:
 $n(Q_1) + n(Q_2) + n(Q_3) + n(Q_4) + 1 = \ell(Q_1) + \ell(Q_2) + \ell(Q_3) + \ell(Q_4) + i(Q_1) + i(Q_2) + i(Q_3) + i(Q_4) + 1$
- Following the rules for the number of vertices $n(Q)$, the number of leaves $\ell(Q)$ and the number of internal nodes $i(Q)$:
 $n(Q) = n(Q_1) + n(Q_2) + n(Q_3) + n(Q_4) + 1$
 $\ell(Q) = \ell(Q_1) + \ell(Q_2) + \ell(Q_3) + \ell(Q_4)$
 $i(Q) = i(Q_1) + i(Q_2) + i(Q_3) + i(Q_4) + 1$
 $\therefore n(Q) = \ell(Q) + i(Q)$

$\therefore n(Q) = \ell(Q) + i(Q)$ holds true for any full quadtree Q ■

2. Prove that for any full quadtree Q , we have $\ell(Q) \leq 4^{h(Q)}$:

- BASIS STEP: For a full quadtree Q consisting of only a root, $\ell(Q) = 1$, and $h(Q) = 0$
- $\therefore \ell(Q) \leq 4^{h(Q)} = 1 \leq 4^0 = 1 \leq 1$ which holds true.
- RECURSIVE STEP: Assume $\ell(Q_1) \leq 4^{h(Q_1)}$, $\ell(Q_2) \leq 4^{h(Q_2)}$, $\ell(Q_3) \leq 4^{h(Q_3)}$, and $\ell(Q_4) \leq 4^{h(Q_4)}$, when Q_1, Q_2, Q_3 , and Q_4 are pairwise disjoint full quadtrees, and r is a vertex not belonging to them, (Q_1, Q_2, r, Q_3, Q_4) , then:
 $\ell(Q) = \ell(Q_1) + \ell(Q_2) + \ell(Q_3) + \ell(Q_4)$
 $h(Q) = 1 + \max(h(Q_1), h(Q_2), h(Q_3), h(Q_4))$
 $\therefore \ell(Q_1) + \ell(Q_2) + \ell(Q_3) + \ell(Q_4) \leq 4^{1 + \max(h(Q_1), h(Q_2), h(Q_3), h(Q_4))}$
 $\ell(Q) \leq 4^{1 + h(Q)}$
 $\ell(Q) = 4 \cdot 4^{h(Q)}$
 $\ell(Q)/4 = 4^{h(Q)}$

Because the height of a quadtree is one less than the number of rows, dividing $\ell(Q)$ by 4 takes away one of these rows to be equivalent to the height.

$\therefore \ell(Q) \leq 4^{h(Q)}$ holds true for any full quadtree Q ■

3. Prove that for any full quadtree Q , we have $n(Q) \leq \sum_{j=0}^h 4^j$ where $h = h(Q)$.

- BASIS STEP: For a full quadtree Q consisting of only a root, $n(Q) = 1$ and $h(Q) = 0$
 $\therefore n(Q) \leq \sum_{j=0}^h 4^j = 1 \leq \sum_{j=0}^0 4^j = 1 \leq 4^0$, which holds ($1 \leq 1$).
- RECURSIVE STEP: Assume $n(Q_1) \leq \sum_{j=0}^{h(Q_1)} 4^j$, $n(Q_2) \leq \sum_{j=0}^{h(Q_2)} 4^j$, $n(Q_3) \leq \sum_{j=0}^{h(Q_3)} 4^j$, and $n(Q_4) \leq \sum_{j=0}^{h(Q_4)} 4^j$, when Q_1, Q_2, Q_3 , and Q_4 are pairwise disjoint full quadtrees, and r is a vertex not belonging to them, (Q_1, Q_2, r, Q_3, Q_4) , then:
 $n(Q) = n(Q_1) + n(Q_2) + n(Q_3) + n(Q_4) + 1$
 $h(Q) = 1 + \max(h(Q_1), h(Q_2), h(Q_3), h(Q_4))$

$$\begin{aligned}
&\therefore n(Q_1) + n(Q_2) + n(Q_3) + n(Q_4) + 1 \leq \sum_{j=0}^{h(Q_1)} 4^j + \sum_{j=0}^{h(Q_2)} 4^j + \sum_{j=0}^{h(Q_3)} 4^j + \sum_{j=0}^{h(Q_4)} 4^j \\
&n(Q_1) + n(Q_2) + n(Q_3) + n(Q_4) + 1 \leq \sum_{j=0}^{1 + \max(h(Q_1), h(Q_2), h(Q_3), h(Q_4))} 4^j \\
&n(Q) \leq \sum_{j=0}^{1 + h(Q)} 4^j
\end{aligned}$$

$\therefore n(Q) \leq \sum_{j=0}^h 4^j$ where $h = h(Q)$, holds true for any full quadtree Q . ■